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### Geometric mask rule check with favorable and unfavorable zones

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#### Abstract

A method includes generating a diffraction map from a plurality of target patterns, generating a favorable zone and an unfavorable zone from the diffraction map, placing a plurality of sub-resolution patterns in the favorable zone, and performing a plurality of geometric operations on the plurality of sub-resolution patterns to generate modified sub-resolution patterns. The modified sub-resolution patterns extend into the favorable zone, and are away from the unfavorable zone.

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## Background/Summary

**PRIORITY CLAIM AND CROSS-REFERENCE** (1) This application is a continuation of U.S. patent application Ser. No. 18/334,551, filed Jun. 14, 2023, and entitled "Geometric Mask Rule Check With Favorable and Unfavorable Zones," which is a continuation of U.S. patent application Ser. No. 17/386,737, filed Jul. 28, 2021, and entitled "Geometric Mask Rule Check With Favorable and Unfavorable Zones," now U.S. Pat. No. 11,714,951, issue Aug. 1, 2023, which claims the benefit of the U.S. Provisional Application No. 63/188,196, filed on May 13, 2021, and entitled "Method for Performing Mask Rule Check with Favorable and Unfavorable Zone," which applications are hereby incorporated herein by reference.

### BACKGROUND

(1) In the formation of lithography masks, which are used for forming patterns for integrated circuits, First Order Diffraction Map (FODM) was used to generate seeds for scattering pattern bars and other Sub-Resolution Assistant Features (SRAFs). The seeds may be modified through relocation, sizing, merging, or separation in order to pass mask rule check (MRC) criteria. This ensures that the patterns meet the manufacturable requirement, such as minimum width, minimum space, minimum area, no acute angle, etc., of the mask-making processes and tools, and hence the lithography masks can be manufactured.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1 through 8 illustrate the intermediate stages in the generation of patterns for a

lithography mask in accordance with some embodiments.

(3) FIGS. **9** through **12** illustrate the cross-sectional views of intermediate stages in the formation of some patterns on integrated circuit component in accordance with some embodiments.

(4) FIG. **13** illustrates a process flow for forming a lithography mask in accordance with some embodiments.

#### DETAILED DESCRIPTION

(5) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(6) Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(7) A method of forming Sub-Resolution Assistance Features (SRAFs) (which are also referred to as scattering bars) is provided. The method includes laying out target patterns that are to be implemented on a wafer, generating a diffraction map from the target patterns, determining favorable zones and unfavorable zones from the diffraction map, generating initial patterns (seeds) in the favorable zones, and enlarging the initial patterns so that the enlarged patterns may pass the minimum width or/and the minimum area constrain of the mask rule checks. Since the enlargement may cause the degradation of optical performance, a pattern modification process is performed to modify the patterns, so that the resulting patterns no longer extend into the unfavorable zone. By keeping the patterns separated from the unfavorable zones, the modified patterns result in a better optical performance, and the target patterns may be implemented better on a photo resist.

Embodiments discussed herein are to provide examples to enable making or using the subject matter of this disclosure, and a person having ordinary skill in the art will readily understand modifications that can be made while remaining within contemplated scopes of different embodiments. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. Although method embodiments may be discussed as being performed in a particular order, other method embodiments may be performed in any logical order.

(8) FIGS. **1** through **8** illustrate the intermediate stages in the generation of patterns for a photo lithography mask in accordance with some embodiments of the present disclosure. The corresponding processes are also reflected schematically in the process flow **200** shown in FIG. **13**.

(9) Referring to FIG. **1**, target patterns **20** are generated/laid out. The respective process is illustrated as process **202** in the process flow **200** as shown in FIG. **13**. Throughout the description, the term “target patterns” refers to the patterns of the target features that are to be implemented on integrated circuit components, which include and are not limited to, device wafers, interposer wafers, package substrates, reconstructed wafers, and the like. The target features may be any of the features that are to be formed, which include, and are not limited to, dielectric regions, semiconductor regions, metallic regions, or the like. Also, the target patterns may be formed on etching masks such as photo resists, which target patterns on the etching masks may then be

transferred to the integrated circuit components.

(10) In order to implement the patterns on the integrated circuit components, the target patterns are to be formed on a photo lithography mask, such as the photo lithography mask **40** shown in FIGS. **9** and **10**. The photo lithography mask includes transparent portions and opaque portions, with either the transparent portions or the opaque portions including the target patterns. The photo lithography mask is used in a lithography process, in which a light beam is projected on the photo lithography mask, so that an underlying photo sensitive material such as a photo resist is exposed. After the exposure and a subsequent development process, the target patterns are transferred into the photo sensitive material, which may then be used as an etching mask to form the target features on the integrated circuit components.

(11) Due to optical effects, especially with the increasing down-scaling of the integrated circuits, the target features may not be implemented on the integrated circuit components accurately. For example, the shapes, the sizes, the spacings, etc., may be distorted. Sub-Resolution Assistant Features (SRAFs) may be used to help to more accurately implement the target features on the integrated circuit components. The SRAFs are formed on the photo lithography masks, and have sizes smaller than the resolution of the respective lithography tool and process. For example, when a 193 nm light beam is used for light exposure with Numerical Aperture (NA) equaling to 0.9 combined with proper illumination shape, the minimum resolution may have pitch equaling to 107 nm and width equaling to about 40 nm, and the features having at least one of lengths and widths smaller than about 20 nm are sub-resolution features. Although the sub-resolution assistant features are formed on the photo lithography masks, the resulting photo resist, after development, will not have these patterns. On the other hand, the target patterns will be formed on the photo resist with improved accuracy due to the help of the sub-resolution assistant features. Alternatively stated, although the sub-resolution assistant features are not on the photo resist, the patterns in the photo resist are closer to the patterns on the photo lithography mask due to the help of the sub-resolution assistant features.

(12) Referring again to FIG. **1**, two example target patterns **20** with square shapes are shown as an example. In actual circuits, however, the patterns may have any shape including, and not limited to, rectangles, hexagons, octagons, circles, ovals, or the like, or combination of these shapes. There may also be a much greater number of target patterns in a circuit. The concept of the embodiments, however, may be explained using simple target patterns.

(13) With target patterns **20** being provided, a diffraction map is generated according to the certain illumination shape which is used for lithography process. The respective process is illustrated as process **204** in the process flow **200** as shown in FIG. **13**. A portion of an example diffraction map **22** is shown in FIG. **2**. Diffraction map **22** includes the distinctive pattern of light and dark fringes, rings, etc., formed due to the diffraction from target pattern **20**. For example, if holes are formed in an opaque plate, and the holes have the shapes and the sizes of the target patterns, when a light beam (with a certain wavelength) is projected on the opaque plate, diffraction map **22** may be formed on another plate behind the opaque plate.

(14) In accordance with some embodiments of the present disclosure, diffraction map **22** is generated through simulation, for example, using a computer with a software configured to simulate the diffraction patterns. The simulation may have different accuracy level depending on the requirement. A more accurate simulation takes longer time to finish, and the resulting simulated refraction map is closer to the actual diffraction map (for example, the one obtained through holes on opaque plates). In accordance with some embodiments, the simulation may be a first-order simulation that with relatively lower accuracy, but takes shorter time to finish. The resulting first-order diffraction map still has some difference from the actual diffraction map, while it may still be accurate enough for implementing the embodiments of the present disclosure. The resulting diffraction map may thus be referred to as a first-order Diffraction Map (FODM) if a first-order simulation is performed. In accordance with other embodiments, the diffraction map may be

generated with higher-order accuracy, and thus may be a second-order diffraction map, third-order diffraction map, or the like. In accordance with yet other embodiments, the diffraction map may be obtained using other methods, such as forming actual patterns on an opaque plate, and projecting a light beam on the opaque plate to obtain the diffraction map directly. All of these methods for generating the diffraction map are in the scope of the present disclosure.

(15) As shown in FIG. 2, diffraction map 22 includes bright patterns 24, which include bright patterns 24A, 24B, 24C, and more, which are not shown. Bright patterns 24A are the patterns of target patterns 20 with distortion caused by optical effect, which is to be corrected by the embodiments of the present disclosure. Bright patterns 24B and 24C are the interference patterns. There may be more bright patterns outside bright patterns 24C. From inner bright patterns 24A to outer bright patterns 24B and 24C, the brightness decreases gradually. The patterns outside of bright pattern 24C may be too dim to distinguish, however. Furthermore, the outer bright patterns are closer to other nearby patterns (not shown), and may be affected by the bright patterns of the nearby pattern. Accordingly, in accordance with some embodiments, bright pattern 24A and 24B, and sometime bright patterns 24A, 24B and 24C, may be adopted by the embodiments of the present disclosure, while the rest of outer bright patterns are ignored in accordance with some embodiments of the present disclosure.

(16) Diffraction map 22 further includes dark patterns 26, which include dark patterns 26A, 26B, 26C, etc., between bright patterns 24. It is appreciated that although FIG. 2 illustrates that there are clear boundaries between bright patterns 24 and dark patterns 26, FIG. 2 is schematic, and in the actual diffraction map, the middle portions of bright patterns 24 are brightest, and the middle portions of dark patterns 26 are darkest. The transition from bright patterns 24 to dark patterns 26 is gradual, and there are no clear boundaries in between.

(17) FIG. 3 illustrates the generation of zone map 32, which includes favorable zone(s) 30 and unfavorable zones 28 (including 28A and 28B) in accordance with some embodiments. The respective process is illustrated as process 206 in the process flow 200 as shown in FIG. 13. It is appreciated that although one favorable zone 30 and two unfavorable zones 28 are shown as an example, the total count of the favorable zone 30 may be any number equal to or greater than one, and the total count of the unfavorable zones 28 may be any number equal to or greater than one. Favorable zones 30 are generated based on bright patterns 24 (FIG. 2), and may include the parts of the bright patterns 24 whose brightness values exceeding a first pre-determined brightness value, which is discussed in subsequent paragraphs. Unfavorable zones 28 are generated based on dark patterns 26 (FIG. 2), and may include the parts of the dark patterns 26 whose brightness values lower than a second pre-determined brightness value, which is discussed in subsequent paragraphs.

(18) Favorable zones 30 are the preferred zones in which the subsequently formed sub-threshold assistant features are to be placed, and the formation of sub-threshold assistant features in these regions help the formation of target features, and reduces undesirable optical effect. Unfavorable zones 28 are the zones in which the placement of the subsequently formed sub-threshold assistant features will worsen the undesirable optical effect. Accordingly, unfavorable zones 28 are also forbidden zones in which the formation of sub-threshold assistant features is forbidden.

(19) In accordance with some embodiments, the determination/generation of favorable zones 30 and unfavorable zones 28 is based on absolute (threshold) brightness values. For example, two brightness values B1 and B2 may be predetermined, with brightness value B2 being greater than or equal to brightness value B1. When the brightness values at certain positions of diffraction map 22 are lower than brightness value B1, the corresponding positions belong to an unfavorable zone 28. When the brightness values at certain positions of diffraction map 22 are higher than brightness value B2, the corresponding positions belong to a favorable zone 30. In accordance with some embodiments, brightness value B1 is lower than brightness value B2. In the resulting zone map, favorable zones 30 are spaced apart from the neighboring unfavorable zones 28, as shown in FIG. 3. The difference (B2-B1) determines the spacings (such as spacings S1 and S2 in FIG. 3) between

neighboring favorable zones **30** and unfavorable zones **28**, and the brightness values **B1** and **B2** determine the widths (such as widths **W1A**, **W1B**, and **W2** in FIG. **3**) of unfavorable zones **28** and favorable zones **30**, respectively. In accordance with alternative embodiments, values **B1** and **B2** are equal to each other. Accordingly, favorable zones **30** touch the corresponding neighboring unfavorable zones **28**. Brightness value **B1** is generally set not to be greater than brightness value **B2** to avoid ambiguity in whether a certain position belongs to an unfavorable zone **28** or a favorable zone **30**.

(20) In accordance with alternative embodiments, the determination of favorable zones **30** and unfavorable zones **28** is based on relative brightness values. It is appreciated that the determination of relative brightness values may have many methods, which are in the scope of the present disclosure. The relative brightness values may be determined based on the highest brightness value of bright patterns, or based on both of the highest brightness value of a bright pattern(s) and the lowest brightness value of a dark pattern(s). In accordance with some example embodiments, the brightness value **B<sub>Brig</sub>** of the brightest point of bright patterns **24** is used as the criteria for generating favorable zones **30** and unfavorable zones **28**. (Threshold) Relative brightness values **F1** and **F2** are also pre-determined, with both of relative brightness values **F1** and **F2** being between, and not including, 0 and 1. In accordance with some embodiments, when the brightness values at certain positions of diffraction map **22** are lower than  $F1 \cdot B_{Brig}$ , the corresponding positions are in unfavorable zones **28**. Conversely, when the brightness values at certain positions of diffraction map **22** are higher than  $F2 \cdot B_{Brig}$ , the corresponding positions are in favorable zones **30**. In accordance with some embodiments, value **F1** is lower than brightness value **F2**. The difference (**F2**−**F1**) determines the spacings (such as spacings **S1** and **S2** in FIG. **3**) between neighboring favorable zones **30** and unfavorable zones **28**, and the relative brightness values **F1** and **F2** determine the widths (such as widths **W1A**, **W1B**, and **W2** in FIG. **3**) of unfavorable zones **28** and favorable zones **30**, respectively. In accordance with alternative embodiments, relative brightness values **F1** and **F2** are equal to each other. Accordingly, favorable zones **30** touch the corresponding neighboring unfavorable zones **28**. Relative brightness value **F1** is generally set not to be greater than relative brightness value **F2** to avoid ambiguity in whether a certain position belongs to an unfavorable zone **28** or a favorable zone **30**.

(21) Referring to FIG. **4**, target patterns **20** and initial Sub-Resolution Assistant Features (SRAFs) **34** are added to the zone map **32**. The respective process is illustrated as process **208** in the process flow **200** as shown in FIG. **13**. Initial SRAFs **34** are also patterns, which are intended to be formed on a photo lithography mask along with target patterns **20**. In the subsequent discussion, initial SRAFs **34** are referred to as scattering bars since they are often formed as having bar shapes. Initial scattering bars **34** have lengths and/or widths smaller than the resolution (hence are sub-resolution features) of the lithography tool and process. Accordingly, scattering bars **34** (even after the subsequent modification) will not be transferred to the resulting integrated circuit components such as wafers, packages, package substrates, etc. . . . As a comparison, target features **20** have lateral dimensions greater than the resolution, and hence their patterns will be transferred to the resulting integrated circuit components. Although the patterns of initial scattering bars **34** are not transferred, their existence on the photo lithography mask affect the optical effect in light-exposure processes, and the transferred patterns on the integrated circuit components are closer in shape and size to the target patterns **20** on the photo lithography mask.

(22) In subsequent discussion, reference numeral **34** is used to represent both of the initial scattering bars and the scattering bars after modification processes. Letter(s) “**M**” and “**MM**” may also be added following reference numeral “**34**” to identify the stage of the modification.

(23) The initial scattering bars **34** have beneficial effect on the transferring of target patterns **20** when initial scattering bars **34** are in bright patterns **24** (FIG. **2**). Accordingly, initial scattering bars **34** are added into favorable zones **30** (FIG. **4**), which are determined based on the brightness of the bright patterns **24**. In accordance with some embodiments, initial scattering bars **34** are rectangular

bars, and some of initial scattering bars **34** may be square bars. In accordance with alternative embodiments, initial scattering bars **34** may have any other shapes including, and not limited to, polygons (such as hexagons, octagons, or the like), circles, ovals, or the like. Initial scattering bars **34** may also have irregular shapes including the combinations of curves, straight lines, and/or the like. Also, an initial scattering bar **34** may be different from, or the same as, another initial scattering bar **34** in shapes, widths, lengths, etc.

(24) In accordance with some embodiments, all of the initial scattering bars **34** are fully inside favorable zones **30**. In accordance with alternative embodiments, some of initial scattering bars **34** may extend slightly out of favorable zones **30**, and no scattering bar **34** extends into unfavorable zones **28**. Scattering bar **34** may be placed suit to the shape and the extending direction of the respective part of favorable zones **30**. Accordingly, some of initial scattering bars **34** may have their lengthwise directions in the X-direction, and some other scatterings bar **34** may have their lengthwise directions in the Y-direction.

(25) Next, a first mask rule check is performed on the patterns that are to be formed into photo lithography masks. The respective process is illustrated as process **210** in the process flow **200** as shown in FIG. **13**. The first mask rule check is geometric-based, and hence may be performed in a short period of time. The first mask rule check ensures that the manufacturing tool for forming photo lithography masks can form the scattering bars **34** and target patterns **20** on photo lithography masks. For example, the patterns that are too small, too close to each other, or having too small areas cannot be formed on photo lithography mask successfully. The mask rule check thus checks the patterns including the initial scattering bars **34** and target patterns **20** to ensure all mask rules are followed. Since the initial scattering bars **34** are formed as sub-resolution features, some of the initial scattering bars **34** may violate mask rules and thus fail to pass the mask rule check. The scattering bars **34** unable to pass mask rule check are referred to as mask-rule violating scattering bars hereinafter. FIG. **4** illustrate some example mask-rule violating scattering bars **34**, which are marked using notations **34F**. The mask-rule violating scattering bars **34F** are also numbered by adding a digit following letter “F” in order to distinguish them from each other.

(26) FIG. **5** illustrates a first scattering bar modification process (also referred to as a mask-rule compliant operation), wherein the mask-rule violating scattering bars **34F** are enlarged to form modified scattering bars **34M** (including **34M1** through **34M6**), so that the modified bars **34M** may pass mask rule check. The respective process is illustrated as process **212** in the process flow **200** as shown in FIG. **13**. Although enlargement is used as an example of the mask-rule compliant operation, the mask-rule compliant operation may also include relocation and/or merging. All of the scattering bars **34** in FIG. **6** are identified using letter “M” to distinguish them from the initial scattering bars in FIG. **4**, regardless of whether these scattering bars are modified or not from the scattering bars shown in FIG. **4**. The first scattering bar modification process is geometric-based. The modification may include enlarging the initial scattering bars **34** either in the X-direction, Y-direction, or in both of the X-direction and the Y-direction. The modification may also include replacing a small scattering bar **34** with a larger scattering bars **34**. FIG. **5** illustrates an example modification process in which the mask-rule violating scattering bars **34F** are extended in the +X and -X directions, while they may also extend in the +Y direction and/or -Y direction.

Furthermore, the mask-rule violating scattering bars **34F** may also be relocated, for example, when the failure in mask rule check is caused by the too-small spacing between the mask-rule violating scattering bars **34F**. The scattering bar modification process may be performed through a software executed in a computer.

(27) The scattering bar modification process is performed for the purpose of passing the mask rule check, while the effect of the scattering bar modification process on the optical performance is not considered. The effect of the modified scattering bars **34M** on the optical performance may be determined through simulation. The simulation, however, takes long time to finish, especially when large integrated circuit components having complicated patterns are simulated. In accordance with



some embodiments of the present disclosure, unfavorable zones **28** are used to at least limit the adverse effect on the optical performance caused by the scattering bar modification process, or improve the optical performance significantly.

(28) FIG. 5 illustrates some examples of the modified scattering bars **34M**. For example, the modified scattering bars **34M1**, **34M2**, **34M3**, **34M4**, and **34M6** extend into (and overlap) unfavorable zone **28A**, and the modified scattering bar **34M5** extends into (and overlap) unfavorable zone **28B**. There may also be some modified scattering bars **34M** that do not extend into unfavorable zones **28A** and **28B** after the first scattering bar modification process.

(29) Since unfavorable zones **28** are determined as including the dark patterns **26** in the diffraction map, when the modified scattering bars **34M** extend into unfavorable zones **28**, the optical performance will be adversely affected. Accordingly, an unfavorable zone check process is performed to identify the scattering bars **34M** that extend into the unfavorable zones **28**. The respective process is illustrated as process **214** in the process flow **200** as shown in FIG. 13. The identified scattering bars **34M** that extend into the unfavorable zones **28** are referred to as unfavorable patterns or unfavorable scattering bars.

(30) When one or more of the modified scattering bars **34M** is identified as unfavorable scattering bars, a second scattering bar modification process is performed to modify unfavorable scattering bars again, and to keep the resulting scattering bars **34** out of the unfavorable zones **28**. The respective process is illustrated as process **216** in the process flow **200** as shown in FIG. 13.

(31) It is appreciated that the unfavorable zone check process **214** is a geometric checking process, which may be performed fast. As a comparison, if the optical performance of the modified scattering bars **34M** are to be determined through simulation, the simulation will take long time. Accordingly, the optical performance determining process in accordance with the embodiments of the present disclosure is much more efficient.

(32) The second scattering bar modification process may include a plurality of geometric-based operations including, and not limited to, shrinking, relocating, merging, removing, and the like, and/or combinations thereof. The resulting modified scattering bars are referred to as **34MM**, which includes **34MM1** through **34MM5**. All of the scattering bars in FIG. 6 are identified as including “MM” to distinguish them from the scattering bars in FIG. 5, regardless of these scattering bars are modified or not from the scattering bars shown in FIG. 5.

(33) In accordance with some embodiments, the modified scattering bars **34M1** (FIG. 5) extends into unfavorable zone **28A**, and is shrunk in a direction away from unfavorable zone **28A** in the second scattering bar modification process. The resulting modified scattering bars **34MM1** (FIG. 6) no longer extends into unfavorable zone **28A**. The modified scattering bars **34M2** and **34M3** (FIG. 5) also extend into unfavorable zone **28A**, and are shrunk in directions away from unfavorable zone **28A**. The resulting modified scattering bars **34MM2** and **34MM3** (FIG. 6) no longer extend into unfavorable zone **28A**. The modified scattering bars **34M4** (FIG. 5) extends into unfavorable zone **28A**, and is merged with scattering bar **34M7** (which may be mask-rule violating or mask-rule compliant) to form a new modified scattering bar **34MM8** (FIG. 6). In accordance with some embodiments, the merging may be achieved by resizing (enlarging or shrinking) and/or relocating one or both of the merged scattering bars **34M4** and **34M7**. The merging causes the merged size to be larger, and hence the merged pattern may pass the minimum area constrain of the mask rule check. The modified scattering bars **34M5** (FIG. 5) extends into unfavorable zone **28B**, and is shrunk in a direction away from unfavorable zone **28B**. The resulting modified scattering bars **34MM5** (FIG. 6) no longer extends into unfavorable zone **28B**. The modified scattering bars **34M6** (FIG. 5) extends into unfavorable zone **28A**, and is removed. FIG. 6 illustrates where the removed scattering bar **34M6** was using a dashed bar.

(34) The second scattering bar modification process may (or may not) be performed with the mask rules taken into account in accordance with some embodiments. For example, the shrinking may be performed so that the resulting shrunk scattering bars **34M1** and **34M2** are still great enough to

pass the minimum width or/and the minimum area constrain of the mask rule check. In accordance with some embodiments, the modification is based on certain pre-determined rules, for example, shrinking to a certain percentage (such as between about 70 percent and about 90 percent) of the original length. The resulting modified scattering bars **34MM** may or may not pass mask rule check in accordance with these embodiments, and further mask rule check, unfavorable zone check, and the corresponding modification processes may be needed.

(35) As the result of the second scattering bar modification process, some or all of the modified scattering bars **34MM1** may be fully inside favorable zones **30**. In accordance with alternative embodiments, some of the modified scattering bars **34MM** are fully inside favorable zones **30**, while some other scattering bars **34MM** (such as **34MM2** and **34MM3**) are partially inside favorable zones **30**, and partially out of favorable zones **30**. These scattering bars **34MM**, however, are outside of unfavorable zones **28**.

(36) Referring to the process flow **200** as shown in FIG. **13**, a rework process may be performed. The rework process may include processes **218** through **224**, and may also include processes **212** and **214** if processes loop back to processes **212** and **214**. The rework process is discussed below.

(37) In accordance with some embodiments, after the second scattering bar modification process **216**, a second mask rule check process may be performed to ensure that the twice-modified scattering bars don't violate mask rules. The respective process is illustrated as process **218** in the process flow **200** as shown in FIG. **13**. For example, the shrinking of scattering bars may cause scattering bars **34MM** to be too small again. If the second mask rule check is passed with no scattering bar **34MM** failing, as shown in process **220** in process flow **200**, the process flow may go to process **222** in FIG. **13**. Otherwise, the process loops back to process **212** in the process flow **200** as shown in FIG. **13**, and another scattering bar enlargement process **212**, the subsequent unfavorable zone check process **214**, scattering bar modification process **216**, etc., are performed again.

(38) In accordance with some embodiments, after passing the second scattering bar modification process, a second unfavorable zone check process **222** may be performed to ensure that the twice-modified scattering bars do not fall into unfavorable zones again. The respective process is illustrated as process **222** in the process flow **200** as shown in FIG. **13**. For example, when relocating a scattering bar away from unfavorable zone **28A**, the scattering bar may extend into unfavorable zone **28B**. The scattering bars that fall into unfavorable zones again will be marked and modified again. As shown in process **224**, if all scattering bars **34** pass the second unfavorable zone check process **222**, the patterns generated in preceding processes may be used to form a photo lithography mask. Otherwise, if one or more scattering bars **34** fail to pass the second unfavorable zone check process **222**, the process loops back to process **216** in the process flow **200** as shown in FIG. **13**, and another scattering bar modification process and the subsequent processes are performed.

(39) FIGS. **7** and **8** illustrate the scattering bars in a rework process as an example. In FIG. **7**, the modified scattering bar **34MM3** extends into unfavorable zone **28B**. Accordingly, another modification process is performed to shrink scattering bar **34MM3**, and to generate scattering bar **34MM3'** as shown in FIG. **8**.

(40) It is appreciated that if the previous processes are improperly performed, rework processes may be performed endlessly. For example, relocating a scattering bar away from unfavorable zone **28A** causes it to extend into unfavorable zone **28B**, and the relocating in the rework may cause it to extend back into unfavorable zone **28B** again. To prevent this from happening, some considerations may be taken into the scattering bar modification process in the rework. For example, assuming a previously modified scattering bar still fails in the unfavorable zone check process **224**, in the resulting re-execution of scattering bar modification process **216**, a new modification operation different from the previous modification operation will be performed. For example, if the previous modification operation was a shrinking operation, the newly performed modification operation may

be relocation, merging, or the like. Alternatively, the same operation may be performed but with different parameters. For example, shrinking rate may be changed from 20% to 15%, or shrinking value is changed from 2.0 nm to 1.5 nm. This may prevent the cyclic operation. In accordance with some embodiments, a pre-determined number of reworks (such as 1, 2, or 3, or more) may be allowed to be performed, with the operations in the reworks different from the previous operations. If there are still scattering bar(s) **34** fail to pass the unfavorable zone check process and/or mask rule check process after the pre-determined number is reached, these scattering bars will be removed to end the loop, or be marked and reported as an error for further handling.

(41) In accordance with some embodiments, if one or more previously modified scattering bar still fail in mask rule check process **220** or the unfavorable zone check process **224**, the failed scattering bars are removed to prevent the further rework, and to prevent cyclic reworks, without further try, or may be marked and reported as an error for further handling.

(42) In above-discussed processes, a two-step scattering bar modification process is performed, which includes enlarging scattering bars **34**, and then performing modification processes, so that the resulting modified scattering bars **34** are kept out of the unfavorable zones. In accordance with alternative embodiments, a one-step scattering bar modification process is performed, wherein in the enlargement of the mask-rule violating scattering bars, the unfavorable zones **28** are considered, and the enlargement is toward selected directions away from the nearest unfavorable zones **28**. The scale of the enlargement in the selected direction is also controlled, so that the resulting enlarged scattering bars will not extend into unfavorable zones **28**. The subsequent mask rule check may be performed or may be skipped. The subsequent unfavorable zone check is no longer needed.

(43) In accordance with some embodiments, as discussed referring to FIGS. **1** through **8**, a pattern-generation process includes generating a diffraction map including both of unfavorable zones **28A** and **28B**, and rework processes are performed. It is appreciated that unfavorable zone **28A** has a greater effect on the optical performance than unfavorable zone **28B**. This means a scattering bar has smaller adverse effect on optical performance extending into unfavorable zone **28B** than extending into unfavorable zone **28A**. Accordingly, in accordance with some embodiments, to improve the efficiency in the generation of patterns, unfavorable zone **28A** is generated, while unfavorable zone **28B** is not generated to compromise efficiency and accuracy. In accordance with these embodiments, rework may be, or may not be performed. In accordance with alternative embodiments, both of unfavorable zones **28A** and **28B** are generated, and no rework process is performed. Alternatively, the scattering bar will be marked and reported as an error for further handling after a certain number (such as 5) of times of iteration of rework.

(44) The target patterns **20** and the scattering patterns **34MM** (FIG. **6** or FIG. **8**) are then used to form a photo lithography mask **40** as shown in FIGS. **9** and **10**. It is appreciated that a photo lithography mask different from what is shown may be used. For example, an extreme Ultraviolet (EUV) mask may be adopted. The respective process is illustrated as process **226** in the process flow **200** as shown in FIG. **13**. FIG. **9** illustrates a top view of a portion of photo lithography mask **40**, in which scattering bars **34MM** (which are also scattering patterns **34**) and target patterns **20** are formed. The favorable zones **30** and unfavorable zones were used to assist the generation of scattering bars **34**, and are not formed in photo lithography mask **40**.

(45) FIGS. **10** through **12** illustrate the cross-sectional view of intermediate stages in the transferring of the patterns in photo lithography mask **40** into integrated circuit component **44** in accordance with some embodiments. Referring to FIG. **10**, lithograph mask **40** includes opaque portions and transparent portions. In accordance with some embodiments, the target patterns **20** and scattering bars **34** are formed as the opaque portions, which are between transparent portions **36**, as shown in FIG. **10**. In accordance with alternative embodiments, target patterns **20** and scattering bars **34** may be formed as transparent portions of the lithograph mask, with opaque portions separating them from each other. FIG. **10** illustrates the reference cross-section **10-10** in FIG. **9**.

(46) Integrated circuit component **44** is placed underneath photo lithography mask **40**. Integrated

circuit component **44** may be a device wafer, an interposer wafer, a package substrate strip, a reconstructed wafer, or the like. Integrated circuit component **44** includes target layer **46**, which may be a dielectric layer, a semiconductor layer, a conductive layer (such as a metal layer), or the like. Photo resist **48** is applied over target layer **46**. Light **50** is projected on lithograph mask **40**, so that photo resist **48** is exposed.

(47) After the light-exposure of photo resist **48**, photo lithography mask **40** is moved away. Photo resist **48** is baked and developed, and some portions are removed. The resulting photo resist **48** includes the patterns of target patterns **20**, but not the patterns of scattering patterns **34**, as shown in FIG. **11**. In a subsequent process, photo resist **48** is used to etch the underlying target layer **46**. The resulting etched target layer **46** again includes the patterns **20** of target patterns **20**, but not the patterns of scattering patterns **34**. Photo resist **48** is then removed, and the resulting structure is shown in FIG. **12**. In the above-discussed processes, scattering patterns **34** result in more accurate transferring of target patterns **20** into target layer **46**, although scattering patterns **34** are not formed in target layer **46**.

(48) In above-discussed processes, the processes shown in FIGS. **1** through **8** may be performed using a computer with software (programming codes) and hardware. The software includes the tools for performing the tasks including, and not limited to, generating (laying out) target patterns, simulating diffraction maps, generating favorable zones and unfavorable zones, generating initial scattering patterns, performing mask rule checks, enlarging scattering patterns, performing unfavorable zone checks, modifying the scattering zones, and the like. The program codes of the software and the results such as the diffraction map, the favorable zones and unfavorable zones, the target patterns and the scattering patterns may be embodied on a non-transitory storage media, such as a hard drive, a disc, or the like, and may be shipped for manufacturing photo lithography masks.

(49) In above-illustrated embodiments, the advanced lithography process, method, and materials described above can be used in many applications, including fin-type field effect transistors (FinFETs). For example, the fins may be patterned to produce a relatively close spacing between features, for which the above disclosure is well suited. In addition, spacers used in forming fins of FinFETs, also referred to as mandrels, can be processed according to the above disclosure.

(50) The embodiments of the present disclosure have some advantageous features. By generating favorable and unfavorable zones, scattering bars are generated, and are separated from unfavorable zones. The optical effect is thus optimized. The optimization of the optical effect is through geometric checking of the scattering bars to decide whether they extend into the unfavorable zones, and hence is fast. This saves the time that is otherwise spent on performing time-consuming simulations to determine the optical effect of the scattering bars.

(51) In accordance with some embodiments of the present disclosure, a method comprises generating a diffraction map from a target pattern, wherein the diffraction map comprises a bright pattern and a dark pattern; generating a favorable zone and an unfavorable zone from the bright pattern and the dark pattern; placing a first plurality of sub-resolution patterns in the favorable zone; performing a mask-rule compliant operation (which may be an enlargement operation, a relocation operation, or a merging operation) on the first plurality of sub-resolution patterns to generate a second plurality of sub-resolution patterns, wherein a first group of sub-resolution patterns in the first plurality of sub-resolution patterns are enlarged; performing an unfavorable zone check process to find unfavorable patterns, wherein the unfavorable patterns are enlarged first group of sub-resolution patterns that extend into the unfavorable zone; and performing a geometric operation on the second plurality of sub-resolution patterns to generate a third plurality of sub-resolution patterns, wherein unfavorable patterns are separated from the unfavorable zone. In an embodiment, the method further comprises a mask rule check process to find the first group of sub-resolution patterns from the first plurality of sub-resolution patterns, wherein the first group of sub-resolution patterns are mask-rule violating patterns. In an embodiment, the first plurality of sub-resolution patterns further comprise a second group of sub-resolution patterns that are mask-rule

compliant, and in the enlargement operation, the second group of sub-resolution patterns are unmodified. In an embodiment, the geometric operation comprises shrinking one of the unfavorable patterns. In an embodiment, the geometric operation comprises relocating one of the unfavorable patterns. In an embodiment, the geometric operation comprises removing one of the unfavorable patterns. In an embodiment, the geometric operation comprises merging one of the unfavorable patterns with another one of the plurality of sub-resolution patterns. In an embodiment, the method further comprises manufacturing a photo lithography mask, wherein the target pattern and the third plurality of sub-resolution patterns are formed in the photo lithography mask; and using the photo lithography mask to form an integrated circuit component, wherein the target pattern is implemented on the integrated circuit component, and the third plurality of sub-resolution patterns are not implemented on the integrated circuit component. In an embodiment, the method further comprises performing a mask rule check process on the third plurality of sub-resolution patterns. In an embodiment, the method further comprises enlarging additional mask-rule violating scattering bars in the third plurality of sub-resolution patterns to generate a fourth plurality of sub-resolution patterns. In an embodiment, the method further comprises performing an additional unfavorable zone check on the fourth plurality of sub-resolution patterns.

(52) In accordance with some embodiments of the present disclosure, a method comprises generating a diffraction map from a plurality of target patterns; generating a favorable zone and an unfavorable zone from the diffraction map; placing a plurality of sub-resolution patterns in the favorable zone; and performing a plurality of geometric operations on the plurality of sub-resolution patterns to generate modified sub-resolution patterns, wherein the modified sub-resolution patterns extend into the favorable zone, and are away from the unfavorable zone. In an embodiment, the diffraction map comprises a bright region and a dark region, and the favorable zone comprises a part of the bright region, and the unfavorable zone comprises a part of the dark region. In an embodiment, the method further comprises determining a first threshold brightness value and a second threshold brightness value equal to or higher than the first threshold brightness value, wherein regions in the diffraction map with brightness values lower than the first threshold brightness value are in unfavorable zones, and wherein regions in the diffraction map with brightness values higher than the second threshold brightness value are in favorable zones. In an embodiment, the plurality of geometric operations comprise an enlargement operation to enlarge some of the plurality of sub-resolution patterns and to generate enlarged patterns; and an additional geometric operation to separate the enlarged patterns from the unfavorable zone. In an embodiment, the method further comprises performing a mask rule check process to find mask-rule violating scattering bars in the plurality of sub-resolution patterns that have gone through some of the plurality of geometric operations. In an embodiment, the method further comprises an unfavorable zone check process to find unfavorable patterns in the plurality of sub-resolution patterns, wherein the unfavorable patterns extend into the unfavorable zone.

(53) In accordance with some embodiments of the present disclosure, a method comprises generating an unfavorable zone and a scattering pattern; determining whether the scattering pattern is overlapped with the unfavorable zone; modifying the scattering pattern to generate a modified scattering pattern, wherein the modified scattering pattern is separated from the unfavorable zone; forming a photo lithography mask comprising the modified scattering pattern; and using the photo lithography mask to perform a light-exposure process on a photo resist. In an embodiment, the method further comprises generating a diffraction map from a target pattern, wherein the target pattern is also in the photo lithography mask; and determining the unfavorable zone and a favorable zone from the diffraction map, wherein the scattering pattern is placed in the favorable zone. In an embodiment, the modifying the scattering pattern comprises enlarging the scattering pattern.

(54) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other

processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

1. A method comprising: generating a diffraction map from target patterns; generating a first plurality of sub-resolution patterns in the diffraction map; selecting a first sub-resolution pattern from the first plurality of sub-resolution patterns, wherein the selecting is based on locations of the first plurality of sub-resolution patterns in the diffraction map; performing a modification operation to modify the first plurality of sub-resolution patterns collectively into a second plurality of sub-resolution patterns, wherein the first sub-resolution pattern is modified to generate a second sub-resolution pattern; and forming a photolithography mask comprising patterns of the target patterns and the second plurality of sub-resolution patterns.
2. The method of claim 1 further comprising determining a first region in the diffraction map as an unfavorable zone, and the first sub-resolution pattern comprises a portion in the unfavorable zone, and wherein the portion of the first sub-resolution pattern in the unfavorable zone is removed to form the second sub-resolution pattern.
3. The method of claim 2, wherein the selecting the first sub-resolution pattern comprises finding one of the first plurality of sub-resolution patterns that extends into the unfavorable zone.
4. The method of claim 2 further comprising determining a second region in the diffraction map as a favorable zone, wherein the first plurality of sub-resolution patterns are generated at least partially in the favorable zone.
5. The method of claim 4, wherein both of the favorable zone and the unfavorable zone are ring-shaped zones, and the unfavorable zone encircles the favorable zone.
6. The method of claim 4, wherein the diffraction map comprises a bright zone and a dark zone, and wherein a first part of the bright zone is determined to be the favorable zone, and a second part of the dark zone is determined to be the unfavorable zone.
7. The method of claim 1, wherein the first plurality of sub-resolution patterns further comprise a second sub-resolution pattern, and wherein both of the first plurality of sub-resolution patterns and the second plurality of sub-resolution patterns comprise the second sub-resolution pattern.
8. The method of claim 7, wherein the second sub-resolution pattern in the first plurality of sub-resolution patterns is identical to the second sub-resolution pattern in the second plurality of sub-resolution patterns.
9. The method of claim 1 further comprising, before the first plurality of sub-resolution patterns are generated: generating a third plurality of sub-resolution patterns in the diffraction map; performing a mask rule check process on the third plurality of sub-resolution patterns; and modifying the third plurality of sub-resolution patterns based on results of the mask rule check process to generate the first plurality of sub-resolution patterns.
10. The method of claim 1, wherein the modification operation comprises an operation selected from the group consisting of shrinking, relocation, merging, and combinations thereof.
11. The method of claim 10, wherein the modification operation comprises merging two of the first plurality of sub-resolution patterns.
12. The method of claim 1, wherein the modification operation further comprises removing an additional sub-resolution pattern from the first plurality of sub-resolution patterns.
13. A method comprising: generating a diffraction map from a plurality of target patterns; determining a favorable zone and an unfavorable zone in the diffraction map, wherein the unfavorable zone encircles the favorable zone; placing a plurality of sub-resolution patterns in the

favorable zone; selecting a first sub-resolution pattern from the plurality of sub-resolution patterns, wherein the first sub-resolution pattern is selected based on a location of the first sub-resolution pattern in the diffraction map; modifying a pattern of the first sub-resolution pattern to generate a modified sub-resolution pattern, wherein the modified sub-resolution pattern is generated based on a second location of the modified sub-resolution pattern; and implementing the plurality of target patterns on a semiconductor wafer.

14. The method of claim 13 further comprising forming the modified sub-resolution pattern and the plurality of target patterns in a lithography mask.

15. The method of claim 14, wherein a second sub-resolution pattern in the lithography mask has an identical pattern as in the plurality of sub-resolution patterns that are placed in the favorable zone.

16. The method of claim 13 further comprising: after the pattern of the first sub-resolution pattern is modified, performing a mask rule check process on the first sub-resolution pattern.

17. The method of claim 13 further comprising modifying the modified sub-resolution pattern.

18. A method comprising: generating a diffraction map from a target pattern; generating a first scattering pattern in the diffraction map; performing a first modification process, wherein the first scattering pattern is modified to generate a modified scattering pattern; after the first modification process, selecting the modified scattering pattern and performing a second modification process to modify the modified scattering pattern and to generate a re-modified scattering pattern, wherein the modified scattering pattern is selected based on a position of the modified scattering pattern in the diffraction map; and forming a photo lithography mask comprising both of the re-modified scattering pattern and the target pattern.

19. The method of claim 18 further comprising determining an unfavorable zone and a favorable zone from the diffraction map, wherein the modified scattering pattern is selected based on a portion of the modified scattering pattern being in the unfavorable zone.

20. The method of claim 19, wherein the portion of the modified scattering pattern in the unfavorable zone is removed to form the re-modified scattering pattern.

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